Question 1

Sumita Khatri wrote a research report and followed the necessary due diligence steps before sharing the report with her client. It turned out that there was a mistake in the report. This was pointed out by her client. Khatri apologized and re-submitted the corrected report. Did Khatri violate any CFA Institute Standards?

A. **No.**
B. Yes, relating to performance presentation.
C. Yes, relating to misconduct.

A is correct. Khatri accepts the mistake and provides the corrected information. Hence she does not violate Standard I (D) – Misconduct. Misconduct requires that members don't engage in any professional conduct involving dishonesty. Since Khatri followed the necessary due diligence steps, she did not violate Standard III (D) – Performance Presentation. *Guidance for Standards I-VII.*

Question 2

Aryana Reid, CFA, is a private wealth manager. She writes a popular blog called "Aryana's Investments" that has several thousand subscribers. The objective of the blog is to attract new clients; every post is also sent as an e-mail to its subscribers. The blog posts are usually a detailed analysis about her investment recommendations and actions. Recently, Reid issued a sell recommendation for Jubilant Inc. However, a few days after publishing her initial recommendation, she decides to change the recommendation from sell to buy based on some new information. In order to comply with the CFA Institute Standards, which of the following is the *most appropriate* method for disseminating the change in investment recommendation?

A. Publish the post and send it as a mail to the blog subscribers.
B. Publish the post, send a mail to blog subscribers and email her clients simultaneously.
C. **Email her clients first.**

C is correct. Standard III (A) Loyalty, Prudence, and Care states that members must place their client's interest before their own interests. Since the intent of the blog is to attract new clients, the change in recommendation must first be communicated to her existing clients. *Guidance for Standards I-VII*

Question 3

Kent Miller is an investment adviser at UBN Investments, who is registered to take Level I of the CFA Exam. He meets with Carrie Hartford, a new client of the firm, who has just moved her account from Northern Investment Bank. She tells Miller that she has read about derivatives and would like to invest in them. Miller explains the upside potential and downside risks of some strategies with derivatives and recommends protective put for her portfolio. To be consistent with the CFA Institute Standards, Miller should:

A. **determine Hartford's needs, objectives, and tolerance for risk before making a recommendation.**
B. explain to Hartford in detail about the characteristics of his firm and the investment vehicles it offers, including the nature of the industry.
C. explain to Hartford his recent candidacy in the CFA Program, and its importance within the investment community.

A is correct. Miller is recommending investment strategies before inquiring about the client's financial situation, investment experience, or investment objectives. Miller is thus violating Standard III (C) – Suitability. B & C provide examples of information that members and candidates should discuss with their clients at the outset of the relationship. But these answers do not constitute a complete list of those factors. Hence, A is the best answer. *Guidance for Standards I-VII*

Question 4

Justin Zoghlin, CFA, was hired as a wealth manager to manage the $2 billion estate of a family in Oklahoma a year ago. He was given the flexibility to choose his working hours. When Zoghlin took the job, he served as the President of his religious community that conducted social welfare programs on a regular basis. In addition, he managed the investments for his large extended family. He did not get paid for his religious community activity or the family investments. Seeing the impressive returns he generated, his friends persuaded him to manage their investments, as well. Now, a year later, he has stopped serving the religious community. He manages investments for non-family members, but charges 10% of the portfolio value as his fee. Zoghlin has not informed his employer of any of these activities. With regard to which of the business activities, has Zoghlin *least likely* violated the CFA Institute Standards of Professional Conduct?

A. **Serving the religious community.**
B. Managing non-family investments.
C. Managing family investments.

A is correct. The activity is not related to financial services. B and C are incorrect as Zoghlin violated Standard IV (B) Additional Compensation Arrangements as there is a potential conflict of interest with the work for his employer. *Guidance for Standards I-VII*

Question 5

Rani Kaporwala, CFA, an analyst at Smart Securities has just finished writing a newsletter to the firm's clients about a new investment strategy involving derivatives. Due to the complex nature of the strategy designed to manage risk and the different trading patterns involved in various economic scenarios, Kaporwala decides to include only the top three liquid securities, with relatively lower volatility in the market to explain the strategy. She consequently withholds information regarding the portfolio construction and valuation scheme. Has Kaporwala violated any CFA Institute Standards?

A. No.
B. **Yes, relating to communications with clients and prospective clients.**
C. Yes relating to fair dealing.

B is correct. Kaporwala has possibly violated the CFA Institute Standard V (B) -Communications with Clients and Prospective Clients, since she needs to inform her clients of the basic process and logic, and the strategy's limitation and inherent risks in case of recommendations involving securities with low liquidity and higher volatility. Without understanding the valuation models, clients will not be able to understand and assess the portfolio construction. *Guidance for Standards I-VII*

Question 6

Cara Knightly is a much respected portfolio manager at RMC investment management company. Knightly is planning to sell her shares in General Tyre Company (GTC) from her personal portfolio to pay for her child's college tuition. Knightly duly informs her supervisor according to the firm's preclearance procedures and receives approval. RMC has however published its research report on GTC with a "buy" recommendation, and Knightly has advised some of her clients to purchase the stock for their portfolios, where suitable. Has Knightly violated any CFA Institute Standards?

A. Yes, related to priority of transactions
B. Yes related to loyalty, prudence, and care.
C. No.

C is correct. Knightly has sought approval in line with the firm's preclearance procedures. She has not violated Standard VI (B) – Priority of Transaction, as it does not limit transactions of company employees that differ from current recommendations as long as the sale does not disadvantage current clients. She did not violate the Standard III (A) – Loyalty, Prudence, and Care since the buy recommendation was made to clients keeping in view their best interests. *Guidance for Standards I-VII*

Question 7
Andy McCarthy, CFA, resigns from his position as a portfolio manager with Drexell Investments and takes up a part-time teaching position at a local college. For the next two years, he does not file a completed Professional Conduct Statement with CFA Institute and does not pay his CFA Institute membership dues. After two years, McCarthy starts his independent practice as a financial analyst and writes his CFA designation on his business cards and next to his name. McCarthy does not reinstate his CFA Institute membership by filing his Professional Conduct Statement and paying the CFA membership fees. He feels that for the past two years he has not been involved in the investment profession therefore he is not required to reinstate his CFA Institute membership. Has McCarthy violated any CFA Institute Standards?

A. No.
B. Yes, relating to reference to the CFA Institute, the CFA designation, and the CFA Program.
C. Yes, relating to loyalty.

B is correct. McCarthy has violated Standard VII (B) – because his right to use the CFA designation was suspended when he failed to file his Professional Conduct Statement and stopped paying dues. Therefore, he is no longer able to state or imply that he is an active CFA charterholder unless he files the Professional Conduct Statement, resumes paying CFA Institute dues and finally completes the CFA Institute reinstatement procedures. *Guidance for Standards I-VII*

Question 8
Heather Dunst, CFA, is the fund manager of a small-cap equity mutual fund. Annabelle Mills, CFA, is responsible for marketing the fund and acquiring new clients. While going through the fact sheet of the fund, Dunst notices that the market value of the total equity holding is lower than the current market value. As a result, the total assets under management is also lower. Mills attributes the difference to the increase in stock price over the past five consecutive sessions. To prevent violation of the CFA Institute Standards of Professional Conduct, Dunst should *least likely:*

A. update the fact sheet with the current market value before sharing it with prospective clients.
B. include a note that states the asset values are as of a particular date.
C. file a report to the CFA Institute's Professional Conduct Program.

C is correct. There is no evidence that any standard has been violated. Hence it does not make sense to file a report to the CFA Institute's Professional Conduct Program. Options A and B represent actions that should be taken. *Code of Ethics and Standards of Professional Conduct.*

Question 9
Pete Yang, CFA, works as an analyst at a large investment management firm. Out of interest, he follows the technology sector closely. He has been following SatCom Inc., a technology company, in particular because of its too-good-to-be true performance of late. After analyzing the stock of SatCom, Yang is convinced the company is resorting to accounting manipulation/fictitious accounts and writes to the capital markets regulator to investigate the company's books. Before he writes to the regulator, Yang shorts the stock of SatCom for his clients' accounts. Subsequently, he also tips the media about the ongoing investigation following which the stock tanks by 20% in a single trading session. Yang covers the short positions, making a good profit for his clients. Which of the following CFA Institute Standards of Professional Conduct did Yang *most likely* violate?

A. Market Manipulation.
B. Priority of Transactions.
C. Preservation of Confidentiality.

A is correct. Yang violated Standard II (B) Market Manipulation by releasing negative information of an ongoing investigation to the public for his clients' benefit.B is incorrect because Standard VI (B) Priority of Transactions has to do with giving personal transactions priority over client or employer transactions, which has not happened here. C is incorrect because SatCom is not a client of Yang. *Guidance for Standards I-VII*

Question 10
Green Meadows Investments is a small brokerage and investment firm which has grown rapidly over the past few years. Howard Grant, owner and CEO, decides to hire a sub-manager to handle the growing demands of the clients to diversify their portfolio by investing in the regional markets. Grant uses the CFA Institute model "request for proposal" to design a questionnaire for the purpose of recruiting a sub-manager. He shortlists five local and regional firms competing for the new business. Grant reviews each one of them in detail and finally hires the most cost-effective firm. Grant has *least likely* violated the CFA Institute Standard relating to:

A. suitability.
B. diligence and reasonable basis.
C. fair dealing.

C is correct. Grant has violated Standard V (A) – Diligence and Reasonable Basis by making his decision on cost alone. He has also violated Standard III (C) – Suitability, since the selected adviser is required to meet the needs of the clients, but Grant has only decided on cost structure, and not ensured whether the new adviser's services are appropriate for his clients. *Guidance for Standards I-VII*

Question 11
Kathy Hughes, CFA, a portfolio manager for high-net-worth clients, is part of a 5-member team that manages a balanced portfolio. Hughes is interviewing for a position at Smart Fund Investments and receives written approval from her current firm to take records of the performance history of the assets she manages. During the interview, she attributes the historical performance of the fund as a collective effort. Hughes also talks at length about specific clients,

their portfolio value, the length of their association, and other success stories. Did Hughes *most likely* violate the CFA Institute Standards by discussing one of the following?

A. **Specific details of the clients.**
B. Discussing the performance history of the fund.
C. Her role in the portfolio's performance.

A is correct. Hughes violated Standard III (E) Preservation of Confidentiality by divulging details of clients constituting the portfolio. B and C are incorrect since she had obtained written approval to discuss the performance history, and she did not misrepresent the performance as her own effort. *Guidance for Standards I-VII*

Question 12
Kavita Karamali is a portfolio manager at Newbridge Investments. One of her clients, Inam Ghauri with a low-risk tolerant profile has asked her to invest in a High-Growth Stock Fund which has shown an average annual return of 23% for the last three years. Ghauri has recently inherited $1.5 million from his aunt. Karamali follows the firm's policy of updating his investment policy statement and considers all the aspects including his investment objectives, constraints, and unique needs. Based on the historical track record, management fees charged, and her client's changed circumstances, she finds the High-Growth Stock Fund as a suitable investment for Ghauri. Nevertheless, Karamali, explains in details all the pros and cons of the investment, the objectives of the Fund pursued by an aggressive investment strategy, finally highlighting the increased risk that Ghauri will be exposed to with the new investment. After obtaining his approval to the terms of the Fund, Karamali invests 30% of his capital in the Fund based on the analysis of the client's new risk appetite and return objectives. The next month the fund announces a loss close to 45% of its value and suspends all redemptions. Ghauri in panic calls the chief executive officer (CEO) of Newbridge and complains about Karamali, questioning her ability as a portfolio manager. The following day Karamali has to defend her decision to invest in the High-Growth Stock Fund for Ghauri, in front of the firm's management committee. Which of the following statements is *most likely* correct?

A. Karamali has violated the Code and Standards by not explaining the Fund's mandate to the client.
B. **Karamali has not violated the Code and Standards.**
C. Karamali has violated the Code and Standards by investing 30% of Ghauri's portfolio in the High-Growth Fund.

B is correct. Karamali's actions were consistent with Standard V (A). Analysis of an investment that results in a reasonable basis for recommendation does not guarantee that the investment has no downside risk. Newbridge should discuss the process of investment and analysis with the client while pointing it out that the past returns do not guarantee future performance, especially in an aggressive strategy fund. *Guidance for Standards I-VII*

Question 13
According to the Fundamentals of the Compliance section of the GIPS standards, which of the following is *most likely* a requirement for claiming compliance?

A. If the firm meets certain but not all requirements of the GIPS standards, it can state that it is "in compliance with the Global Investment Performance Standards except for ...." to indicate partial compliance with the GIPS standards.
B. **Firms must make every reasonable effort to provide a compliant presentation to all prospective clients.**

C. Firms must provide a complete list of composite descriptions to existing clients but not to any prospects that make such a request.

B is correct. According to the requirements of the Fundamentals of Compliance, firms must make every reasonable effort to provide a compliant presentation to all prospective clients. Firms must not choose to whom they present compliant presentation. As long as a prospective client has received a compliant presentation within the previous 12 months, the firm has met this requirement. *The GIPS Standard*

Question 14
Rohan Sharma is an independent advisor based in Mumbai. All his clients are domiciled in India. The Securities and Exchange Board of India (SEBI) launched REITs recently to make investments in the real estate sector accessible to investors without buying physical property. To promote REITs as an asset class, advisors are given incentives to sell REITs. Rohan perceives REITs as a good diversification tool and recommends it to all his clients. Rohan least likely violated which of the following standards?

A. Independence and Objectivity.
B. Suitability.
C. Knowledge of the law.

C is correct. There is no violation of law as a REIT is permitted as a product in India. He is most likely in violation of Standard I (B) Independence and Objectivity as he has an incentive to sell REITs. He also violated Standard III (C) Suitability by recommending it to all his clients. Guidance for Standards I-VII

Question 15
Jayesh Shaan, CFA, is a portfolio manager at Kiaan Investment Company. He develops a derivatives selection model, with promising simulated performance results. He wants to apply the model in managing the risk of his clients' portfolios. He writes a newsletter to all his clients and prospects, explaining the basic format of the model outlining its risks and limitations. He then summarizes the performance results over the past five years using both the historical actual annual outcome of portfolios and the corresponding simulated result. Shaan writes the following concluding statement, "By developing and applying the proprietary derivatives' selection model, the return potential of certain portfolios i.e. where appropriate to include derivatives, increases by 100 basis points over the gross actual returns, and the simulated risk is lowered by 20-30 basis points from the actual risk." "The model is worth employing but Kiaan Investment Company does not guarantee future returns based on past results." Has Shaan violated any CFA Institute Standards?

A. Yes, relating to misrepresentation.
B. Yes, relating to communication with clients and prospective clients.
C. No.

C is correct. Shaan does not violate any CFA Standards of Professional Conduct in his letter. He presents fair, accurate, and complete information when he presents both actual and simulated results in accordance with Standard III (D) – Performance presentation. He does not guarantee superior future investment returns, in accordance with Standard I(C) – Misrepresentation, and describes the basic format of the model giving risks and limitations of the new investment model as per Standard VI (B) – Communications with clients and prospective clients. Guidance for Standards I-VII

Question 16
Carla Santiago is a portfolio manager. She receives a call from a brokerage firm, Amer Securities. Amer Securities encourages Santiago to route her client trades through their firm. As an incentive, Santiago is told that she will receive a discount on her personal trades. Santiago will most likely be in violation of which CFA Institute Standards if she trades through Amer Securities?

A. Diligence and Reasonable Basis.
B. Loyalty, prudence and care.
C. Responsibilities of supervisors.

B is correct. Santiago will most likely violate the CFA Institute Standard III (A) – Loyalty, Prudence and Care if she trades through Amer Securities. Since, Amer Securities will trade Santiago's personal account on favorable terms, she is placing her own interests before those of her clients. Guidance for Standards I-VII

Question 17
A former wealth manager, Jack Hughes, has decided to start a private money management firm, Western Capital. Hughes believes that the new firm needs to quickly build assets under management of €1.50 billion by the end of its first financial year in order to be successful. Therefore, Hughes uses public records to contact former clients to solicit their business. He also advertises on various web portals with the following statement: "Great opportunities and incentives of up to €15,000 for investment advisers who join the firm with a minimum of €150,000 in committed investments." Which of the following is most likely correct according to the CFA Institute Standards?

A. A member or candidate may not accept Hughes' offer because the committed investments of €150,000 required from the investment advisers violate the Code and Standards.
B. A member or candidate who is currently employed may accept the new position, because he/she can arrange for enough clients to move with him/her to the new firm.
C. A member or candidate currently unemployed is eligible for the new position.

C is correct. A & B are inconsistent with the Code and Standards. C is correct because Standard IV (A) – Loyalty discusses activities permissible to members and candidates when they are leaving employment. Soliciting clients is strictly prohibited. Guidance for Standards I-VII

Question 18
Ollie Green is a research analyst with Stellar Investments, and has been asked to write a report on Royal Tento Inc. (RTI), one of the leaders in the mining industry. During a recent mining and metallurgical event organized by the mining industry, he learns that RTI has won the contract for iron ore mining at an extremely attractive price due to the faith placed by that country's government in RTI's ability and reputation to explore and produce successfully in far flung territories. Green meets with the representative of RTI at the event who confirms the news. Back at the office, Green digs up more about RTI and its competitors and also downloads various independent research reports on RTI and the industry. He decides on a "buy" recommendation for RTI stock by stating, "It's a fact that the price of iron ore is on an upward trajectory, hence future prospects of RTI are very bright." Before Green presents his report to his supervisor for review, he instructs his portfolio manager to purchase RTI's stock for his portfolio. The CFA Institute Standard least likely violated by Green is:

A. **diligence & reasonable basis.**
B. priority of transactions.
C. communication with clients and prospective clients.

A is correct. Green least likely violated the Standard V (A) – Diligence and Reasonable Basis, since his recommendation was based on an extensive research. Green has violated Standard VI (B) – Priority of Transactions, by trading ahead of his report being published. He called his portfolio manager to buy shares for himself before publishing the recommendation, putting his interests before the client's interests. Green also failed to differentiate fact from opinion regarding the price of iron ore in his report and therefore violated the standard regarding communication with clients. Guidance for Standards I-VII

Question 19
Hassan Sooriya recently left his job as a portfolio manager at an investment bank to start his own investment management firm. Few years ago, Sooriya had earned his CFA charter, which he feels should be advertised in the new marketing material he is creating for his firm. He makes the following two statements in the marketing material:
I. "As a CFA charterholder, I am committed to the highest ethical standards.
II. The CFA Program is globally recognized and attests to a charterholder's success in the field of investment management and analysis."
Has Sooriya violated Standard VII (B) – Reference to CFA Institute, the CFA Designation, and the CFA Program in his advertising material?

A. No.
B. Sooriya violated Standard VII (B) in the first statement.
C. **Sooriya violated Standard VII (B) in the second statement.**

C is correct. Sooriya violated Standard VII (B) – Reference to CFA Institute, the CFA Designation, and the CFA Program in his second statement. Proper reference would have been, "The CFA designation is globally recognized and attests to a charterholder's success in a rigorous and comprehensive study program in the field of investment management and research analysis." Guidance for Standards I-VII

Question 20
Survivorship bias refers to

A. Showcasing only the top-performing portfolio to represent the firm's overall investment results.
B. **Excluding poorly performing portfolios and presenting an average performance history.**
C. Presenting performance for a selected time-period during which the mandate produced excellent returns.

B is correct. A pertains to representative accounts and C refers to varying time periods which were other misleading practices observed in the absence of GIPS. *Introduction to the Global Investment Performance Standards (GIPS)*

Question 21
Chris Isaac, works as a registered broker and investment advisor with TopWorth Securities and Brokerage. Isaac has just attended a rigorous training session organized and led by the Stacy Fenworth, regional director of TopWorth and the compliance department to educate and train the

brokers of the firm about a new comprehensive fee-based account, GoldAccount. At the end of the training session, Fenworth had distributed compliance procedures for establishing and maintaining GoldAccount which is specifically for clients with assets over USD100,000. She had also explained to the participants that as per regulatory rules, they should have reasonable basis for assigning the account to a particular customer. An annual review of the GoldAccount will be conducted to ascertain its status and applicability. Isaac feels that Tracy Keegan, one of his clients who is an active trader, with assets now approaching to USD110,000, is the right candidate since her transaction fees will be discounted by nearly 75% from their current costs. Keegan agrees after reading the disclosure material provided by Isaac and accepts the recommendations. Has Isaac violated any CFA Institute Standards by recommending the GoldAccount to Keegan?

A. Yes, related to diligence and reasonable basis.
B. Yes, relating to suitability
C. No.

C is correct. Isaac does not violate the CFA Institute Standards when recommending the GoldAccount to Keegan. His actions comply with Standard III (A) Loyalty, Prudence and Care, Standard III (C) Suitability and Standard V (A) Diligence and Reasonable Basis. Keegan agrees after reading the relevant disclosures of the account. Based on her asset value and frequent trading, the GoldAccount is suitable for her. Guidance for Standards I-VII

Question 22

Stuart Anderson, CFA, is a portfolio manager at Maxima Investments. He manages fee-paying client accounts including accounts of his close family members. The family members' accounts are similar in every way to the regular clients' accounts and pay the same fee. When the eagerly awaited IPO of SK&L Pharmaceuticals becomes available, Anderson first distributes the issue to his family members, then to his clients, provided it is appropriate for them and finally to his own account. Which of the following standards has Anderson least likely violated?

A. Suitability.
B. Priority of transactions.
C. Fair dealing.

A is correct. The text says 'close family members', which implies that Anderson probably benefits if the family members do well. In this case he should give other clients preference over accounts where he is the beneficial owner. Hence it is possible that Anderson has violated Standard VI (B) – Priority of Transaction. He has clearly violated Standard III (B) – Fair Dealing. This is because he gave his family members preference over other clients. He has least likely violated Standard III (C) – Suitability. Guidance for Standards I-VII

Question 23

Gary Sanders is the head of compliance of a 1,000-employee asset management firm. He has been diagnosed with cancer and is going on a 4-month medical leave. He has always ensured the compliance policies are up-to-date and rigorously enforced within the organization. Before leaving, Sanders hands over his supervisory responsibilities to Kirsten Gates and states his responsibilities include ensuring compliance with all applicable laws, rules, regulations and firm policies. Sanders violated the CFA Institute Standards of Professional Conduct by not including which of the following as part of his supervisory responsibilities?

A. Preventing violations of the CFA Institute Code and Standards.
B. Conducting training programs on compliance.

C. **Both A and B.**

C is correct. Members and candidates with supervisory responsibility must also make reasonable efforts to detect violations of Code and Standards. Furthermore, training programs are important to continually educate personnel regarding the compliance procedures.

Question 24
Anil Poddar, CFA, is an equity research analyst specializing in the banking sector. A team of four analysts, including Poddar analyzed the rights issue of the State Bank of India, India's largest public sector bank. Based on their research, they issued a buy recommendation. Though Poddar agrees with the team on most of the points, he strongly disagrees with them on one aspect: the bank's rising NPAs and no steps taken by the management to curtail it. He feels the stock is overpriced and should not be a buy. Despite Poddar's view, the research report is circulated with his name to all the clients. In order to comply with the CFA Institute Standards, which of the following actions by Poddar would be the least appropriate?

A. Leave the report as is with his name on it.
B. **Release a new report with the sell recommendation.**
C. Remove his name and dissociate from the report.

B is correct. Releasing a separate sell recommendation will confuse clients and cause embarrassment to the firm. Poddar may either choose to remove his name from the report or choose not to dissociate from the report even if he does not agree with some of its content. Guidance for Standards I-VII

Question 25
Timothy Urban works as an independent financial planner. He hires independent professionals for several associated services such as tax planning and filing, insurance, legal, and brokerage services. Each service professional bills Urban for the services provided based on the volume of work they receive from him on a monthly basis. Some of the service professionals pay Urban a referral fee for recommending clients. To comply with the CFA Institute Standards of Professional Conduct, which of the following steps must Urban most likely take?

A. Invoice clients immediately after the independent professionals bill Urban for the service offered.
B. **Keep potential clients informed about his arrangement and fee structure before they sign an agreement with him.**
C. Disclose details of referral fees to clients upon request.

B is correct. As per Standard VI (C) – Referral Fees, potential clients must be informed of the benefit received before they enter into a formal agreement for service, and not after the event. Guidance for Standards I-VII

Question 26
According to the Fundamentals of Compliance section of the GIPS standards, which of the following is *least likely* a recommendation that firms are encouraged to implement?

A. **GIPS Reports should be provided to all existing clients.**
B. Non-discretionary portfolios must not be included in a firm's composites.
C. Firms should adopt the broadest and most meaningful definition of the firm.

A is correct. Firms should provide GIPS Reports to all prospective clients and prospective pooled fund investors. Options B and C are listed under the recommendations. *Introduction to the Global Investment Performance Standards (GIPS)*

Question 27
Cassandra Turnbull is planning a college fund for her daughter who will start college after exactly 6 years. The expected college expense will be USD55,000 per year payable at the start of each year for four years. Turnbull plans to invest in a fund which yields 6%. Turnbull has USD100,000 in savings at this time. What equal amount must she invest at the end of year 1, year 2, year 3, year 4 and year 5 to fund her daughter's four-year college?

A. **$10,069.**
B. $18,116.
C. $24,000.

A is correct. N = 4, I/Y = 6, PMT = –55,000, FV = 0, CPT PV. PV = 190,580.81. Note: this is the amount needed one year before the first payment of 55,000. Cassandra needs $190,580.81 at the end of year 5. Now we have to calculate the equal payments which need to be made at the end of every year. N = 5, I/Y = 6, PV = 100,000, FV = –190,580.81, CPT PMT. PMT = 10,068.7 She needs to save an additional 10,068.7. *Time Value of Money; Section 6.1.*

Question 28
A decrease in the price of a normal good and an increase in the purchasing power of consumers, will cause its demand to:

A. increase due to income effect.
B. increase due to substitution effect.
C. **increase due to both substitution and income effects.**

C is correct. When the price of a normal good decreases, its demand will increase due to both substitution and income effects. *Topics in Demand and Supply Analysis*; Section 2.5.

Question 29
An analyst observes the following data for Rodríguez Company (in millions of **€**):

	2005(€)
Sales	54.6
Cost of goods sold	42.0
Purchases	40.1
Average receivables	15.2
Average payables	13.4
Average inventory	8.9

The company's cash conversion cycle (in days) is *closest* to:

A. **57.**
B. 98.
C. 137.

A is correct. CCC = DSO + DOH – Days Payables = (365 / (54.6 / 15.2)) + (365 / (42 / 8.9)) – (365 / (40.1 / 13.4) = 101.61 + 77.345 – 121.97 = 57. *Financial Analysis Techniques; Section 4.3.*

Question 30
An equity analyst has categorized some stocks based on the industry sector to which they belong. Which measurement scale is the analyst *most likely* using?

A. Interval.
B. Nominal.
C. Ordinal.

B is correct. Data is categorized, but not ranked under a nominal scale. Under an ordinal scale data is ranked, while under an interval scale, data is ranked and separated by equal intervals. *Organizing, Visualizing, and Describing Data*

Question 31
The following data apply to a country in its domestic currency unit:

Consumer spending on goods and services	986,070	Government spending on goods and services	416,700
Business gross fixed investment	397,500	Government gross fixed investment	95,230
Change in inventories	-79,600	Capital consumption allowance	9,650
Transfer payments	9,400	Statistical discrepancy	-3,960
Exports	329,900	Imports	360,990

Using the expenditures approach, the country's gross domestic product (GDP) is *closest* to:

A. 1,771,200.
B. 1,780,850.
C. 1,790,500.

B is correct. Using the expenditures approach:
GDP = Consumer spending on goods and services + Business gross fixed investment + Change in inventories + Government spending on goods and services + Government gross fixed investment + Exports – Imports + Statistical discrepancy
i.e. GDP = 986,070 + 397,500 – 79,600 + 416,700 + 95,230 + 329,900 – 360,990 – 3,960 GDP = 1,780,850. *Aggregate Output, Prices and Economic Growth; Section 2.3.*

Question 32
Steve Rowling is analyzing historical performance of a portfolio manager who claimed to outperform its benchmark 70 percent of the time over a one-year time horizon. Steve has eight months record of the portfolio manager's monthly performance. The portfolio manager outperformed the benchmark in five months. If all the outperformance results are independent and the portfolio manager's performance is as claimed, what is the probability of observing two or fewer profitable recommendations out of eight in total?

A. 7.5%
B. 5.76%

C. 1.13%

C is correct. The observed success rate is 5/8 = 0.625, or 62.5 percent. The probability of two or fewer successes is $F(2) = p(2) + p(1) + p(0)$, where $p(2)$, $p(1)$, and $p(0)$ are respectively the probabilities of 2, 1, and 0 successes, according to the binomial distribution with $n = 8$ and $p = 0.70$. We have
$P(x) = P(X = x) = (n\ x)\ p^x(1-p)^{n-x} = n! / (n-x)!x!\,p^x(1 – p)^{n-x}$
$p(2) = (8!/2!6!)(0.70^2)(0.30^6) = 28\ (0.000357) = 0.009996$
$p(1) = (8!/1!7!)(0.70^1)(0.30^7) = 8\ (0.000153) = 0.001224$
$p(0) = (8!/0!8!)(0.70^0)(0.30^8) = 1\ (0.000066) = 0.000066$
Summing all these probabilities, you conclude that $F(2) = 0.009996 + 0.001224 + 0.000066 = 0.011286$, or 1.13%. The probability of observing two or fewer profitable recommendations out of eight in total is 1.13%. *Common Probability Distributions; Section 2.2.*

Question 33
The market demand function for chairs is given by the equation:
$Q_D = 80 – 0.6\ P + 0.05\ I$
Where Q_D is the chairs demanded per month (in thousands), P is the average price of a chair (in USD) and I is the household monthly income (in thousand USD). The price of a chair is USD25 and monthly income of a family is USD65,000. The own-price elasticity of demand for chairs is *closest* to:

A. -0.6105.
B. -0.2307.
C. -0.2198.

C is correct. The own-price elasticity of demand is given by $(\Delta Q_D/\Delta P)\ (P/Q)$ and notice from the demand function that $\Delta Q_D/\Delta P = −0.6$. Inserting the given variable values into the demand function yields $Q_D = 80 – 0.6 \times 25 + 0.05 \times 65 = 68.25$. (Note: The monthly income is in thousand USD. Therefore we enter 65 in the formula) So at a price of 25, the own-price elasticity of demand equals (–0.6) (25/68.25) = −0.2198. *Topics in Demand and Supply Analysis; Section 2.2*

Question 34
An economist is evaluating the relation between economic growth and currency appreciation. He uses historical data to determine the probability that the economy grows is 63%, the probability that the economy shrinks is 24% and the probability that the economy stays flat is 13%. He later finds out that the currency appreciates. The economist estimates that given the new information regarding the currency, the probabilities that the economy, shrinks or stays flat are as follows:

- P(currency appreciation | economy grows) = 74%
- P(currency appreciation | economy shrinks) = 11%
- P(currency appreciation | economy unchanged) = 15%

What is the probability that the economy will shrink, given the new information that the currency appreciated?

A. 5.16%
B. 11.0%.
C. 23.5%

A is correct. According to Bayes' Theorem: P(E|I) = P(E) x P(I|E) / P(I). Here the event is that the economy shrinks. The information is that the currency appreciates. P(I) = [(0.74 × 0.63) + (0.15 ×

0.13) + (0.11 × 0.24)] = 0.51. Plugging into the formula we have: P(economy shrinks | currency appreciates) = 0.24 x 0.11 / 0.51 = 0.0516. *Probability Concepts; Section 4.1.*

Question 35
The marginal propensity to consume in an economy is 73 percent and the tax rate is 32 percent. Given that planned expenditures are to increase by USD8 billion, the increase in total income and spending (USD in billions) is *closest* to:

A. **USD15.9.**
B. USD11.8.
C. USD29.6.

A is correct. The fiscal multiplier is 1/ [1 – c*(1-T)] = 1/ [1 – 0.73 (1-0.32)] = 1.986. With government expenditure of USD8 billion, total income and spending will rise by USD8 billon * 1.986 = USD15.89 billion. *Monetary and Fiscal Policy; Section 3.2.*

Question 36
Which of the following companies will have the *lowest* financial reporting quality?

A. **A company that combines the results of the two segments it operates in: healthcare and retail.**
B. A company that reports good performance due to favorable exchange rate movement.
C. A company that provides delayed reports, but ones that are GAAP-compliant and decision-useful.

A is correct. Assessing the profitability of each segment is difficult if they are combined. This is an example of biased accounting choice. *Financial Reporting Quality; Section 2.*

Question 37
Bernhard Schwaiger, an equity analyst, wants to look at the ROE of all the companies in the S&P 500 for the year 2015. He would *most likely* require:

A. time-series data.
B. **cross-sectional data.**
C. panel data.

B is correct. Data on some characteristics of companies at a single point in time is cross-sectional data. Organizing, Visualizing, and Describing Data

Question 38
An analyst is analyzing the impact of regulatory intervention in order to subdue speculative behavior on the stock market of Country A. For this purpose, he is studying the return volatility in the stock exchange of Country A before and after the events. He is using the 120 months of return data before and after August 2006. The level of significance is 1%.

	n	Mean Monthly Return (%)	Variance of Returns
Before August 2006	120	1.498	20.855
After August 2006	120	1.402	12.605

Which of the following statement is *most likely* correct?

A. The null hypothesis will be tested using a chi-square test statistic of 1.654 and would conclude that null hypothesis can be rejected.
B. The null hypothesis will be tested using a F test statistic of 0.604 and would conclude that null hypothesis cannot be rejected.
C. The null hypothesis will be tested using a F test statistic of 1.654 and would conclude that null hypothesis can be rejected.

C is correct. The null hypothesis consistent with the verbal description of the research goal would be $\sigma^2_{\text{before}} = \sigma^2_{\text{after}}$.

The alternative hypothesis consistent with the verbal description of the research goal would be $\sigma^2_{\text{before}} \neq \sigma^2_{\text{after}}$.

The "before" sample variance is larger, so F statistic will be 20.855 / 12.605 = 1.654. Because this a two-tailed test, we use F-tables for the 0.005 level (i.e. 0.01 / 2) to give a 0.01 significance level. In the table, the closest value to 119 degrees of freedom is 120 degrees of freedom. At the 0.01 level, the rejection point is 1.61. Because 1.654 is greater than critical value of 1.61, we can reject the null hypothesis that the population variance of returns is the same in the before and after periods.

A is incorrect because to test a null hypothesis of the equality of two variances, we use an F-test. B is incorrect because F-statistic is 1.654 not 0.604. *Hypothesis Testing; Section 4.2. LO. j.*

Question 39
A separate purchases account is most likely required by:
A. the periodic inventory system.
B. the perpetual inventory system.
C. the specific identification valuation method.

A is correct. The periodic inventory system requires a separate purchases account. Inventories; Section 3.6.

Question 40
Gareth Mike is a treasury analyst at Charterhouse Investments Management Company. Mike is reviewing the pricing of a bond of Company Xerrom Inc. The bond is speculative-grade, zero-coupon bond with one-year-maturity. The probability that the bond will default is 5%. The one-year US Treasury bills (T-bills) are offering a return of 2.75% (denoted as R_F). The return (yield to maturity) of a bond is denoted as R. Assuming that the bondholders will recover nothing in the event of a default, which of the following is most likely correct statement?
A. The investor will require a default risk premium of at least 5.41% on the bond to break even compared with investing in the T-bill.
B. The investor will require a default risk premium of at least 3% on the bond to break even compared with investing in the T-bill.
C. The investor will require a default risk premium of at least 5.00% on the bond to break even compared with investing in the T-bill.

A is correct. The two events that may affect returns of an investor and how they will affect the bond value are as follows:
Bond defaults Bond does not default
Bond value USD0 USD(1+R)
The expected value of the bond (per USD1 invested) is calculated as
E(bond) = USD0 × P(the bond defaults) + USD(1 + R)[1 – P(the bond defaults)]
So, E(bond) = USD(1 + R)[1 − P(the bond defaults)] The expected value of the T-bill per $1 invested = (1 + R_F)
In fact, this value is certain because the T-bill is risk free. The next step requires economic reasoning. You want the default premium to be large enough so that you expect to at least break even compared with investing in the T-bill. This outcome will occur.
For calculating the yield to maturity of a bond, we will use the following equation:
Expected value of Bond = Expected value of T-bill
USD(1 + R)[1 − P(the bond defaults)] = (1 + R_F)
Solve for R -> R = {(1 + R_F/[1 − P(the bond defaults)]} – 1 = [1.0275/(1 − 0.05)] − 1 = 1.0816 − 1 = 0.0816 or about 8.16%, and default risk premium is R − R_F = 8.16% − 2.75% = 5.41%.
Probability Concepts; Section 2. LO. l.

Question 41
An analyst is interested in interpreting the results of and performing tests of hypotheses for the market model estimation that regresses the return on a small cap stock against the return on the small cap index over the last six years. He has generated the regression results presented in Exhibits 1 and 2:

Exhibit 1

Statistics	**Change in Index (X)**	**Stock Returns (Y)**
Sum	59.80	50.48
Sum of squared deviations from the mean	19.27	38.21
Sum of cross-products of deviations from the mean	22.91	

Exhibit 2

Source	**Sum of Squares**	**Degrees of Freedom**	**Mean Square**
Regression	13.7348	1.0000	13.7348
Error	5.5401	4.0000	1.3850
Total	19.2749		
Coefficient of Determination	71.26%		
Standard Error of the Estimate	1.1769		
Slope coefficient	0.5995		
Intercept	4.92		

Critical t-values for a 0.05 level of significance:

One-sided, left side: − 2.1318

One-sided, right side: +2.1318

Two-sided: ± -2.7764

Which of the following statements is *least* accurate?

A. The model would be best described as time-series regression.
B. The scatter plot summarizing this relation would be most likely upward sloping.
C. The estimated slope coefficient is not different from zero at the 0.05 level of significance.

C is correct because the estimated slope coefficient is different from zero at the 0.05 level of significance. To determine if slope coefficient is different from zero at the 0.05 level of significance, the t-statistic is calculated by subtracting the hypothesized population slope (B_1) from the estimated slope coefficient () and then dividing this difference by the standard error of the slope coefficient,
t = b^1 – B_1 / Sb^1 = Estimated Value – Hypothesized Value / Standard Error
The standard error of the slope coefficient is calculated as the ratio of the model's standard error of the estimate (SEE) to the square root of the variation of the independent variable:
The standard error of the slope coefficient is calculated as the ratio of the model's standard error of the estimate (SEE) to the square root of the variation of the independent variable

$$s_{\widehat{b_1}} = \frac{SEE}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}}$$

$$= \frac{1.1769}{\sqrt{38.21}} = 0.1904$$

$$t = \frac{\widehat{b_1} - B_1}{s_{\widehat{b_1}}} = \frac{0.5995 - 0}{0.1904} = 3.1491$$

For a 5% significance level, the decision is to reject the null hypothesis because the calculated t-statistic of 3.1491 is greater than 2.776. There is sufficient evidence to indicate that the slope is different from zero. Introduction to Linear Regression

Question 42
Assume that the nominal spot exchange rate (USD/AUD) increases by 6.5 percent, the Australian price level decreases by 2.5 percent, and the U.S. price level increases by 2 percent. The change in the real exchange rate is *closest* to:

A. 1.80 percent.
B. 1.82 percent.
C. 1.14 percent.

A is correct. Here AUD is the base currency. The real exchange rate = nominal exchange rate * price level in AUD / price level in USD. Assume that initially the nominal exchange rate = 1, the price level in AUD = 1 and the price level in USD = 1. Hence the real exchange rate = 1. After the changes the real exchange rate = [(1 + 0.065) * (1 – 0.025)] / (1 + 0.02) = 1.0180. This represents a change of 1.80% relative to the initial value of 1. *Currency Exchange Rates; Section 2.*

Question 43
A desirable attribute of an unbiased estimator is that it should have the least variance among other unbiased estimators of the same population. This attribute is *most likely* known as:

A. consistency.
B. efficiency.
C. unbiasedness.

B is correct. An unbiased estimator is efficient if no other unbiased estimator of the same parameter has a sampling distribution with smaller variance. *Sampling and Estimation; Section 4.1.*

Question 44
A firm will *most likely* shut down its production in the short-run, when:

A. the total revenue is less than the total cost.
B. the total revenue is less than the total variable cost.
C. the marginal revenue is equal to the marginal cost.

B is correct. If the total revenue is less than the total variable cost, the firm will shut down production in the short run and exit the market in the long run. But if the total revenue is enough to cover the total variable cost and not the total fixed cost, the firm will stay in the market in the short run. Option C (marginal revenue = marginal cost) is a profit maximizing condition. *Topics in Demand and Supply Analysis; Section 3.2.*

Question 45
Pardew Jewels had 340,000 common shares outstanding and 45,000 shares of preferred stock paying USD10 dividend per share. Each preferred share is convertible into one common share. In 2008, the company reported a net income of USD2.1 million for the year. What is the reported diluted EPS on the company's financial statements?

A. 4.85
B. 5.45
C. 6.18

A is correct.
Diluted earnings per share = Net Income /(Weighted Number of Shares Outstanding + New Common Shares)
Diluted Earning Per Shares = 2,100,000 / (340,000 + 45,000) = 5.45
Basic Earning Per Shares = Net Income – Preferred Dividends / (Weighted Number of Shares Outstanding)
Basic Earning Per Shares = (2,100,000 – 450,000) / 340,000 =4.85
Since the diluted EPS exceeds the basic EPS, the reported diluted EPS is equal to basic EPS.
Understanding Income Statements; Section 6.3.

Question 46
An investment analyst has data on 80 potential companies to include in a stock portfolio. She has employed three screening criteria for inclusion of stocks in her portfolio with the number of companies meeting the criteria provided below:

Screening Criterion	Number of Companies meeting the screen
P/E Ratio > 5	25
Return on Equity > 20%	33
Current Ratio > 1	48

Assuming that the screening criteria are independent, the probability that a given company will meet all three screening criteria is *closest* to:

A. **8%.**
B. 15%.
C. 24%.

A is correct. The joint probability of the three independent criteria is calculated as: = (25/80) x (33/80) x (48/80) = 0.3125 x 0.4125 x 0.6 = 0.077 or 7.7% ≈ 8% of the 80 companies. *Probability Concepts; Section 2.*

Question 47
Which of the following statements about Giffen goods is least accurate?

A. Demand curve is positively sloped.
B. Income effect overwhelms the substitution effect.
C. **Income and substitution effects are in the same direction.**

C is correct. The income effect overwhelms the substitution effect, so an increase in the price of a Giffen good results in greater demand, hence resulting in a positively sloped (upward sloping) demand curve. Topics in Demand and Supply Analysis; Section 2.6.

Question 48
The following information is available about an economy which produces only two products, K and M:

	Product K		Product M	
Year	**Output (units)**	**Selling price/unit**	**Output (units)**	**Selling price/unit**
2014	5,000	$60	3,600	$34
2015	4,000	$72	3,100	$28

If the implicit price deflator for GDP in 2014 was 100, the GDP deflator for 2015 is closest to:

A. **108.5.**
B. 92.2.
C. 122.3.

A is correct.

	Nominal GDP	Real GDP
2014	5,000 * 60 + 3,600 * 34 = 422,400	422,400
2015	4,000 * 72 + 3,100 * 28 = 374,800	4,000 * 60 + 3,100 * 34 = 345,400

GDP deflator = (Nominal GDP / Real GDP) * 100
GDP deflator = value of current output at current prices/ value of current output at base year prices * 100
GDP deflator = (374,800 / 345,400) * 100 = 108.5. *Aggregate Output, Prices and Economic Growth; Section 2.1.*

Question 49
Martin is planning to pay his child's college tuition for 5 years starting 10 years from now. The current annual cost of college is USD50,000, and this cost is expected to rise at an annual rate of 3 percent. It is assumed that he can earn 3.5 percent annually. How much must he put aside each year, starting next year, if he plans to make 9 equal payments?

A. 24,866
B. 31,007
C. 27,405

B is correct. The payments in year 10, 11, 12, 13, and 14 are the future values of a lump sum of $50,000 in Year 0.
With r = 3%, future value of a lump sum is calculated as $FV_N = PV (1 + r)^N$ for 5 years.
$FV_{10} = 50,000 (1.03)^{10} = 67,196$
$FV_{11} = 50,000 (1.03)^{11} = 69,212$
$FV_{12} = 50,000 (1.03)^{12} = 71,288$
$FV_{13} = 50,000 (1.03)^{13} = 73,427$
$FV_{14} = 50,000 (1.03)^{14} = 75,629$
Using the formula for the PV of a lump sum (r = 3.5%), equate the 5 college payments to single payments as of t = 9, and add them together
$67,196 (1.035)^{-1} + 69,212 (1.035)^{-2} + 71,288 (1.035)^{-3} + 73,427 (1.035)^{-4} + 75,629 (1.035)^{-5} =$ USD321,496
Now solve for payments = X as follows:
N = 9
%i = 3.5%
PV = 0
FV = USD321,496
PMT compute = 31,007.
He needs to put aside $31,007 each year if he starts next year and make 9 equal payments.
Time value of money. Section 4.1, 6.3. LO.e, f.

Question 50
For a lump sum investment of USD125,000 invested at a stated annual rate of 5.5% compounded semi-annually, the number of years needed to grow the sum to USD950,000 is closest to:

A. 57 years.
B. 68 years.
C. 37 years.

C is correct. The effective annual rate (EAR) is calculated as follows:
EAR = (1 + Periodic interest rate)^m – 1
EAR = (1 + 0.055/2)^2 – 1
EAR= (1.055756) – 1 = 0.055756 ≈ 5.58%.
Solving for N on a financial calculator results in (where FV is future value and PV is present value):
(1 + 0.0558)N = FV/PV = USD950,000/USD125,000 è N = 37.35 years. *Time Value of Money. Section. 3.3. LO. c.*

Question 51
A company recently purchased a warehouse property and related equipment for €20 million, which were valued by an appraiser as follows: land €6 million, building €9 million, and equipment €8 million. The company incurred the following additional costs in getting the warehouse ready to use:

- €1.0 million for repairs to the building's roof and windows which will extend the life of the building
- €0.3 million to modify the interior layout to meet their needs
- €0.2 million on an orientation and training session for employees to familiarize them with the facility

The cost to be capitalized to the building account (in millions) for accounting purposes is *closest* to:

A. €20.0.
B. €21.3.
C. €24.5.

B is The capitalized cost of the building would include the purchase price plus the other costs that are directly attributable to the building and are involved in extending its life or getting it ready to use: 20 + 1 + 0.3 = 21.3. *Long Lived Assets; Section 2.1.*

Question 52
Larson Inc. is involved in manufacturing consumer products and has series of outlets which are being operated through contractual agreements separately. One of the Companies with which it has contractual agreement is GoTo. All inventory risk and credit risk on all goods which are being sold via these outlets is borne by Larson. Larson pays commissions to GoTo. Prices are being used as set by Larson Inc. During 2019, Larson had the following information:

- Total sales price of items sold by Larson through GoTo during 2019 was USD4,750,000.
- Total commissions paid by Larson to GoTo during 2019 for these items was USD800,000.

Which of the following statement (s) is most likely correct with respect to the above stated information?

A. GoTo should report $4,750,000 revenue on its 2019 income statement.
B. GoTo should report $800,000 revenue on its 2019 income statement.
C. GoTo should report $3,950,000 revenue on its 2019 income statement.

B is correct. In this example, Larson is acting as a Principal because it is primarily responsible for all inventory risk or credit risk and have discretion in setting the price; whereas GoTo is acting as an agent. Because GoTo is acting as an agent it should report revenue equal to its commission of $800,000 on its income statement. Understanding Income Statements; Section 3.1. LO.b.

Question 53
Aggregate demand (AD) curve will shift left if:

A. interest rates are lower.
B. taxes are higher.
C. companies are operating at near or full capacity.

B is correct. Higher taxes lead to lower disposable income which results in lower consumption. Furthermore, this leads to lower investment spending by businesses.
A is not correct because AD curve will shift right if interest rates are lower as money supply increases. Investment is higher. Higher income will result in higher expenditure. Consumers hold real money balances.
C is not correct because AD curve will shift right if companies are operating at near or full capacity. This will lead to increase in investment spending.

Question 54
The following information is from a company's investment portfolio:

Investment

Classification	Measured at cost or amortized cost
Market value, 31 Dec 2009	10,000
Cost/Amortized cost 31 Dec 2009	12,000
Market value, 31 Dec 2010	9,000
Cost/Amortized cost 31 Dec 2010	10,000

If the investment is reclassified as measured at fair value through other comprehensive income as of 31 December 2010, the balance sheet carrying value of the company's investment portfolio would *most likely*:

A. remain the same.
B. decrease by 1,000.
C. decrease by 2,000.

B is correct. Investments measured at fair value through profit or loss and measured at fair value through other comprehensive income are carried at market value, whereas investment measured at cost or amortized costs are carried at amortized cost. If the investment is reclassified as measured at fair value through other comprehensive income in 2010, the carrying amount should be adjusted to its market value, which is USD9,000. Compared with the amortized cost of USD10,000, it's a decrease of USD1,000. *Understanding Balance Sheets; Section 4.5.*

Question 55
An analyst runs a cross-sectional regression for 100 companies, where the dependent variable is the annual return on stock and the independent variable is the net profit margin (NPM). The results of this simple linear regression estimation are shown in the following Exhibit:

Exhibit: Regression Statistics

Source	**Sum of Squares**	**Degrees of Freedom**	**Mean Square**
Regression	365.3450	1.0000	365.3450

Error	1,725.1110	98.0000	17.6032
Total	2,090.4560		

After studying the results of the regression, the analyst arrives at the following conclusions. Which of the following statements is *least* accurate?

A. The coefficient of determination of the regression model is close to 17.48%.
B. Standard error of the estimate is 3.3037.
C. At 5% level of significance, with critical F-value =3.938, there is sufficient evidence to indicate that that the slope coefficient is statistically different from 0.

B is correct. Calculations will be as follows:
Standard error of estimate (SEE) = $MSE^{1/2}$ = $17.6032^{1/2}$ = 4.1956
Coefficient of determination (R^2) = RSS / SST = 365.345/2090.456 = 17.48%
F = MSR / MSE = 365.345 / 17.6032 = 20.7545
The F-statistic tests whether all the slope coefficients in a linear regression are equal to zero. Since the calculated F-stat is higher than the critical value of 3.938, we can conclude that the slope coefficient is statistically different from 0. *Introduction to Linear Regression.*

Question 56
Assume that the nominal spot exchange rate (USD/EUR) increases by 7.2%, the eurozone price level decreases by 3%, and the U.S. price level increases by 2%. The change in the real exchange rate (%) is closest to:

A. 12.72%.
B. 1.94%.
C. -2.52%.

B is correct. Change in the real exchange rate = [(1 + Change in exchange rate) × (1 + Change in price level in foreign country)] / (1 + Change in price level in domestic country) – 1 = [(1.072)(0.97)] / (1.02) – 1 = 0.0194 = 1.94%. Currency Exchange Rates; Section 2.

Question 57
The most *appropriate* way to account for the assets, that have been selected to be spun-off, until the distribution occurs is to classify them as:

A. held for sale with no depreciation taken.
B. held for use until disposal with no deprecation taken.
C. held for use until disposal with depreciation continuing to be taken.

C is correct. Long-lived assets that will be disposed of other than by sale, such as a spin-off, an exchange for other assets, or abandonment, are classified as held for use until disposal and continue to be depreciated until that time. *Long Lived Assets; Section 4.*

Question 58
The joint probability of returns, for securities X and Y, are as follows:

	Return on security Y=10%	*Return on security Y=5%*
Return on security X = 20%	0.8	0
Return on security X = 10%	0	0.2

The covariance of the returns between securities X and Y is *closest* to:

A. 4
B. 8
C. 12

B is correct. Expected return on security X = 0.8 × 20% + 0.2 × 10% = 18%
Expected return on security Y = 0.8 × 10% + 0.2 × 5% = 9%
Cov (X, Y) = 0.8 [(20 – 18)(10 – 9)] + 0.2 [(10 – 18)(5 – 9)] = 8. *Probability Concepts; Section 3.*

Question 59
An analyst is evaluating the balance sheet of Company T that uses the LIFO accounting method for inventory. Company T prepares its financial statements under US GAAP. The analyst collects the following data:

	31 Dec 15
Inventory as per balance sheet	USD500,000
LIFO reserve	USD70,000
Average tax rate	30%

To compare Company T with a company which uses FIFO, at 31 December 2015, the inventory should be adjusted to:

A. USD479,000
B. USD549,000
C. USD570,000.

C is correct. FIFO inventory = LIFO inventory + LIFO reserve
FIFO inventory = $500,000 + $70,000 = $570,000. Inventories; Section 4.1.

Question 60
A country's international transactions accounts data for last year are presented in its domestic currency:

Transaction	**Amount**
Exports of goods and services	1,000
Import of goods and services	1,400
Investment income payments made to foreigners	250
Investment income received from foreigners	340
Net change in assets owned abroad	150
Net change in foreign-owned assets domestically	490
Unilateral current transfers received	34
Unilateral current transfers paid	110
Statistical discrepancy	64

The current account balance is *closest* to:

A. -322.
B. -386.
C. -450.

B is Correct.

Transaction	Amount	Totals
Export of goods and services and income receipts		
Export of goods and services	1,000	
Investment income received from foreigners	340	
		1,340
Import of goods and services and income payments		
Import of goods and services	-1,400	
Investment income payments made to foreigners	-250	
		-1,650
Net unilateral current transfers		
Unilateral current transfers received	34	
Unilateral current transfers paid	-110	
		-76
Current account balance		-386

International Trade and Capital Flows; Section 4.2.

Question 61
If the expected return on other assets fall, the speculative demand for money will:
A. increase.
B. decrease.
C. remain unaffected.

Question 62
The financial ratios of Devon Barton Company are given below:

Operating profit margin	22.4%
Net profit margin	16.7%
Dividend payout ratio	60%
Return on assets	13.1%
Debt to equity	0.68

The company's sustainable growth rate is closest to:

A. 3.6%.
B. 8.8%.
C. 13.2

B is correct. Financial leverage = (0.68 + 1) ÷ 1 = 1.68
ROE = ROA × Financial leverage = 13.1 × 1.68 = 22%
Sustainable growth rate = Retention ratio × ROE = (1 – 0.6) × 22% = 8.8%. Financial Analysis Techniques; Section 5.2.

Question 63
An analyst calculates the following ratios for two companies:

	Alpha Inc.	Beta Inc.
Net profit margin	9.6%	8.5%
Debt/EBITDA	2.8%	2.1%
Retained cash flow to debt	5.4%	7.3%
Free cash flow to net debt	1.3%	4.8%

Which company will *most likely* be assigned a higher credit rating?

A. Alpha Inc.
B. Beta Inc.
C. Cannot say.

B is correct. Beta Inc. has a higher retained cash flow relative to debt and higher cash flow to net debt. Cash flows are more relevant to debt repayment capacity than profit margins. *Applications of Financial Statement Analysis; Section 4.*

Question 64
An analyst gathers the following information to determine the relationship between two variables:

Value of the test statistic	1.94
Critical value at the 0.05 significance level	1.96
Critical value at the 0.01 significance level	2.58

If the null hypothesis is that the population correlation coefficient is equal to zero and alternative hypothesis is that the population correlation coefficient is different from zero, then the analyst *most likely* conducted a:

A. one-tailed test and can reject his null hypothesis.
B. two-tailed test and can reject his null hypothesis.
C. two-tailed test and cannot reject his null hypothesis.

C is correct. Because the alternative hypothesis is that the correlation is different from zero (either above or below zero), the analyst conducted a two-tailed test. Because the test statistic is lower than either of the critical values, the analyst cannot reject the null hypothesis. *Hypothesis Testing; Section 2.*

Question 65
For 2013, Judy Co. reported cost of goods sold of USD25 million, beginning inventory of USD2 million, and an ending inventory of USD5 million. If the accounts payable increased by USD3 million, the cash paid to its suppliers during the year is closest to:

A. $22 million.

B. **$25 million.**
C. $28 million.

B is correct. Cash paid to suppliers = COGS + Increase in inventory — Increase in accounts payable = USD25 million + 3 million – USD3 million = USD25 million. Understanding Cash Flow Statements; Section 3.1.

Question 66
Which of the following statements related to goodwill is most likely correct?
A. If a company is not making new acquisitions, goodwill remains constant.
B. Goodwill is amortized over time.
C. **Goodwill is tested for impairment annually.**

Solution: C is correct. Under both IFRS and US GAAP, accounting goodwill arising from acquisitions is capitalized. Goodwill is not amortized but is tested for impairment annually. If goodwill is deemed to be impaired, an impairment loss is charged against income in the current period. An impairment loss also reduces total assets i.e. goodwill can change even if the company is not making new acquisitions. Understanding Balance Sheets; Section 2.

Question 67
Ryan Ali is an investment advisor at Global Money Investment Company. His client has asked him to recommend a portfolio comprising of 2 banking stocks, 3 utility stocks, and 1 oil stock. If Ryan can choose from 4 banking stocks, 5 utility stocks, and 4 oil stocks, how many different sets of 6 stocks are possible?
A. 60
B. **240**
C. 2,880

B is correct. Ryan needs to form a different combination of 6 different stocks.
- 2 banking stocks out of 4 banking stocks can be selected in 4C2 = 6 ways
- 3 utility stocks out of 5 utility stocks can be selected in 5C3 = 10 ways
- 1 oil stock out of 4 oil stocks can be selected in 4C1 = 4 ways

Thus, the total ways to select 7 stocks = 6 × 10 × 4 = 240 ways. *Probability Concepts. Section. 4.2. LO. o.*

Question 68
Which of the following assets should most likely be tested for impairment at least annually?
A. Factory equipment.
B. A patent expiring in 8 years.
C. **A trademark with an indefinite life.**

C is correct. Intangible assets with indefinite lives need to be tested for impairment at least annually. PP&E and intangibles with finite lives are only tested if there has been a significant change or other indication of impairment. Long Lived Assets; Section 5.

Question 69
Primark wrote down the value of its inventory in 2018 and reversed the write-down in 2019. Compared to the results the company would have reported if the reversal had never occurred, Primark's reported 2019:

A. cost of sales was overstated.
B. cash flow from operations was understated.
C. Taxes were understated.

B is correct. The reversal of the write-down shifted cost of sales from 2019 to 2018. The 2018 cost of sales was higher because of the write-down, and the 2019 cost of sales was lower because of the reversal of the write-down. Hence, due to reversal, 2019 profit was higher, which led to higher taxes and lower cash flow from operations.
If the reversal had never occurred, the company's 2019 cost of sales was higher, profit was lower and accordingly, taxes were lower and cash flow from operations was higher. *Inventories. Section 7.2. LO. h.*

Question 70
McMillan Corp enters a contract which requires payments of USD57 million at the end of each of the next five years for exclusive use of a specific machine over that time period. The present value of the payments is close to USD234 million. At the end of the contract, McMillan will return the machine. The contract does not contain a purchase option. The machine can be used in many applications by many types of customers. The remaining useful life of the machine is ten years, and its fair value is USD235.6 million. This contract is:

A. an operating lease.
B. an unsecured loan.
C. a finance lease.

C is correct. The contract is a finance lease. One of the five criteria for classifying a lease as a finance lease is met. The present value of the lease payments equals substantially all of the fair value (234/235.6= 99%). *Non-Current (Long-Term) Liabilities*

Question 71
The following information is available for Fernando Oil Ltd.

Net income	USD65,000
Depreciation	USD16,000
FC investment	USD8,000
WC investment	USD12,000
Net debt repayment	USD36,000
Cash flow from operations	USD69,000

The free cash flow to equity of the company is *closest* to:

A. USD13,000.
B. USD25,000.
C. USD29,000.

B is correct. FCFE = CFO – FC Investment – Net debt repayment = 69,000 – 8,000 – 36,000 = $25,000. *Understanding Cash Flow Statements; Section 4.3.*

Question 72
Which of the following is *least likely* a lagging indicator for the U.S. economy?

A. **Industrial production index.**
B. Inventory-sales ratio.
C. Unit labor costs.

A is correct. Industrial production index is a coincident indicator. *Understanding Business Cycles; Section 5.*

Question 73
Which feature for preparing financial statements pertains to important information not being omitted?

A. Fair presentation
B. **Materiality and aggregation**
C. No-offsetting

B is correct. Materiality and aggregation pertain to avoiding omission of important information. Similar information should be grouped together.
Fair presentation relates to faithful representation of transactions.
No offsetting means assets and liabilities, and revenue and expenses should not be offset against each other.

Question 74
An analyst uses the following data to estimate the effects of the changes in the INR exchange rate on India's balance of trade.

	Volume (INR billions)	Demand elasticity
Exports	250	0.7
Imports	300	0.9

A depreciation of the INR will most likely:

A. **reduce the trade deficit.**
B. increase the trade deficit.
C. have no effect on the trade deficit.

A is correct. If the Marshall-Lerner condition is satisfied, a depreciation of the domestic currency will reduce an existing trade deficit.
$\omega_X\varepsilon_X + \omega_M (\varepsilon_M - 1) > 0$.
250/550 x 0.7 + 300/550 x -0.1 = 0.318 – 0.054 = 0.264 > 0. Currency Exchange Rates; Section 5.1.

Question 75
TedFin Inc. issued $32 million of fixed rate unsecured bonds at an interest rate of 9.5%. The rates have decreased to 8.75% since then. Using only the carrying amount (based on historical cost) reported on the balance sheet to analyze the company's financial position, which of the following statement is most likely correct?

A. the economic liabilities of TedFin Inc. may be overestimated
B. the economic liabilities of TedFin Inc. may be underestimated.
C. the changes in a liability's fair value will be reported in other comprehensive income

B is correct. When interest rates fall, the fair value of a bond with a fixed coupon rate increases. Thus, the carrying amount of the bonds being carried on the balance sheet is lower than the market value. As a result, a company's economic liabilities may be higher than its reported debt based on amortised historical cost. So the economic liabilities are underestimated. Because the bonds are issued at a fixed rate, there is no effect on interest coverage.
C is incorrect because a company electing to measure a liability at fair value will report decreases in the liability's fair value as income and increases in the liability's fair value as losses. Because changes in a liability's fair value can result from changes in market rates and/or changes in the credit quality of the issuing company, accounting standards require companies to present separately the portion of the change resulting from changes in their own credit risk. Specifically, the company will report the portion of the change in value attributable to changes in their credit risk in other comprehensive income. Only the portion of the change in value not attributable to changes in their credit risk will be recognised in profit or loss. *Non-Current (Long-Term) Liabilities; Section .3. LO. b.*

Question 76
According to the IASB Conceptual Framework, the qualitative characteristic of useful financial information pertaining to the agreement of different knowledgeable users on the information presented to be a faithful representation of the economic events as intended, is *best* described as:
A. comparability.
B. understandability.
C. verifiability.

C is correct. Under the International Accounting Standards Board's Conceptual Framework, verifiability is the qualitative characteristic that means that different knowledgeable and independent users would agree that the information presented faithfully represents the economic events that it is intended to represent. *Financial Reporting Standards; Section 5.*

Question 77
Kim Norton is analyzing the average monthly income of the working population in a small town. She takes a sample of 250 working individuals and found a sample mean of USD3,265 with a sample standard deviation of USD884. Given that Z_{025} = 1.96 and $Z_{0.05}$ = 1.645, there is 95% probability that the population mean is between:
A. USD1,532.4 and USD4,997.6.
B. USD3,155.4 and USD3,374.6.
C. USD3,209.1 and USD3,320.9.

B is correct. The 95% confidence interval uses z_{025} as the reliability factor. The cumulative probability value closest to 0.975 provides the appropriate value of $z_{0.025}$ which is $z_{0.025}$ = 1.96. The confidence interval = 3265 ± 1.96 x (884/√250) = USD3,155.4 and USD3,374.6. *Sampling and Estimation; Section 4.2.*

Question 78
A secondary school student is equally likely to score more than 10, and up to 25 marks in a math test. Given that the marks are only awarded in integers, the probability that the student scores more than 19 marks, and up to 23 marks, is *closest* to:

A. 0.250.
B. 0.267.
C. 0.333.

B is correct. Total possible outcomes = 15. Probability of each outcome = 1/15 = 0.0667.
P $(19 < x \leq 23)$ = P (x = 20, 21, 22, 23)
P $(19 < x \leq 23) = 4 \times 1 \div 15 = 0.2667$. *Probability Concepts; Section 2.*

Question 79
Different depreciation methods for accounting and tax purposes led to the following differences in the carrying amount and tax base for Hobart Petroleum's oil filtration plant.

	Carrying amount	Tax base
End of the year 2013	USD7.5 million	USD8.0 million
End of the year 2014	USD7.0 million	USD7.2 million
End of the year 2015	USD7.7 million	USD6.4 million

The plant was revalued in December 2015, with a revaluation surplus of $1.2 million. The applicable tax rate is 35%. The deferred tax liability on 31 December 2015 is *closest to*:

A. USD35,000.
B. USD420,000.
C. USD455,000.

A is correct. Deferred tax liability = (carrying value – tax base) × tax rate = (7.7 million – 6.4 million) × 35% = 455,000. The effect of revaluation must be removed as it goes directly to equity. 455,000 – (1,200,000 × 35%) = 35,000. *Income Taxes; Section 2.*

Question 80
In which of the following cases is collusion least likely to occur?

A. When the product is homogeneous.
B. When the market shares of the companies are similar.
C. When the cost structures of companies are similar.

B is correct. Collusion is least likely to occur when companies have similar market shares because then competitive forces tend to outweigh the benefits of collusion. The Firm and Market Structures; Section 5.1.

Question 81
Company A is offered a five-year lease contract for use of an asset with an implied interest rate of 7% and an annual lease payment of EUR 75,000 per year payable at the end of each year. Therefore, the present value of the lease payments is close to EUR 307,515. The asset will be amortized over the five-year lease term on a straight-line basis. Company A reports under IFRS. In Year 3, interest expense and amortisation expense respectively will be closest to:

A. **EUR 13,778 and EUR 61,503**
B. EUR 13,778 and EUR 61,222
C. EUR 9,492 and EUR 65,508

A is correct. In Year 3, interest expense is EUR 13,778 and amortization expense is EUR 61,503 as illustrated in the following tables:

Year	Lease Payment	Interest Expense	Principal Repayment	Lease Liability
0				307,515
1	75,000	21,526	53,474	254,041
2	75,000	17,783	57,217	196,824
3	75,000	13,778	61,222	135,601
4	75,000	9,492	65,508	70,093
5	75,000	4,907	70,093	0
Total	**375,000**	**67,485**	**307,515**	

Year	Amortisation Expense	ROU Asset
0		307,515
1	61,503	246,012
2	61,503	184,509
3	61,503	123,006
4	61,503	61,503
5	61,503	0

Non-Current (Long-Term) Liabilities

Question 82
At the beginning of the year, a company had total shareholders’ equity consisting of ¥500,000 in common share capital and ¥100,000 in retained earnings. Following details are given for the year (in ¥):
Net income reported 35,000
Dividends paid 3,500
Unrealized loss on available-for-sale investments 1,000
Repurchase of company stock, to be held as treasury stock 5,000
The total shareholders’ equity at the end of the year is closest to:

A. **¥625,500.**
B. ¥626,500.
C. ¥631,500.

A is correct.

Shareholders' equity = 500,000 + 100,000 + 35,000 – 3,500 – 1,000 – 5,000 = 625,500. Understanding Balance Sheets; Section 6.2.

Question 83
A monopolist faces the following demand and cost schedules:

Output (units)	Price ($/unit)	Total costs ($)
0	8,000	600
15	7,700	35,600
30	7,400	78,600
45	7,100	104,600
60	6,800	197,600
75	6,500	281,600

The optimal output level for this producer is closest to:

A. 75.
B. 15.
C. 45.

C is correct. The optimal price level is 45 units because it produces the highest profit.

Output (units)	Price ($/unit)	Total revenue ($)	Total cost ($)	Profit
15	7,700	115,500	35,600	79,900
30	7,400	222,000	78,600	143,400
45	7,100	319,500	104,600	214,900
60	6,800	408,000	197,600	210,400
75	6,500	487,500	281,600	205,900

Topics in Demand and Supply Analysis; Section 3.2.

Question 84
A company which prepares its financial statements in accordance with IFRS issues £2,000,000 face value five-year bonds on January 1, 2013 when interest rates are 5.60%. The bonds carry a coupon of 4.70%, with interest paid annually on December 31. The carrying value of the bonds as of December 31, 2014 will be closest to:

A. £1,824,960.
B. £2,049,295.
C. £1,951,527.

C is correct. The bond's carrying value is determined by taking the present value of the cash flows over the remaining 3 years. N = 3, I/Y = 5.6, PMT = 2,000,000 x 4.7% = 94,000, FV = 2,000,000. Compute PV = 1,951,527. Non-Current Liabilities; Section 2.2.

Question 85
Three countries produce leather and cotton. The following table illustrates the output per worker per day in each country:

Country	Leather	Cotton
A	7	9
B	5	8
C	3	6

Which country is most likely to have the greatest comparative advantage for producing leather?

A. **Country A.**
B. Country B.
C. Country C.

A is correct. A country has a comparative advantage if its opportunity cost of producing a product is less than the opportunity cost of its trading partners.

Country	Leather	Cotton	Comparative advantage (cotton/leather)
A	7	9	1.29
B	5	8	1.60
C	3	6	2.00

International Trade and Capital Flows; Section 2.4.

Question 86
Which ratio is most likely higher for a company using FIFO method to account for inventory, during a period of rising prices, when compared against a company using weighted average cost method?

A. Debt-to-equity ratio.
B. Inventory turnover.
C. **Return on sales.**

C is correct. In periods of rising prices, FIFO results in a higher inventory value and a lower cost of goods sold and therefore a higher net income. The higher net income increases return on sales. The higher reported net income also increases retained earnings, and therefore results in a lower debt-to-equity ratio not a higher one. The combination of higher inventory and lower cost of goods sold decreases inventory turnover (COGS/inventory). Inventories; Section 7.2.

Question 87
An analyst has gathered the following information about an economy:

Total population	15,000
Working age population	12,250
Labor force	11,950
Underemployed	4,500

Unemployed	3,540
Discouraged workers	2,340
Frictionally unemployed	2,250
Voluntarily unemployed	2,000

The unemployment rate in the economy is closest to:

A. **29.6 percent.**
B. 23.6 percent.
C. 28.9 percent.

A is correct. Unemployment rate = (Unemployed / Labor force) * 100 = (3,540 / 11,950) * 100 = 29.6%. Understanding Business Cycles; Section 4.1.

Question 88

A company pays prepaid rent which is immediately tax deductible, but for accounting purposes, the prepaid expense is not recognized until the rent has been utilized. It is most likely that the company will report:

A. **a deferred tax liability.**
B. a deferred tax asset.
C. no deferred tax asset or liability.

A is correct. A deferred tax liability is created since the prepaid rent (asset) exceed the tax base (which is now zero). A deferred tax liability arises when the carrying value of an asset exceeds its tax base. Income Taxes; Section 2.

Question 89

Solo Bank NA follows US GAAP to report its financial statements. Some information about deferred taxes is given below:

	2011	**2010**
Deferred tax asset	$22,698	$18,650
Valuation allowance	$7,450	$6,822

The change in the valuation allowance most likely reflects that for Solo Bank:

A. the expectations of future earning power have increased.
B. **the expectations of future earning power have decreased.**
C. the deferred tax liabilities were increased in 2011.

B is correct. The valuation allowance is taken against deferred tax assets to represent the uncertainty that future taxable income will be sufficient to fully recover the assets. An increase in valuation allowance indicates greater probability that future earnings will not be offset by the deferred tax asset. Financial Reporting Quality; Section 4.2.

Question 90

To avoid selecting stocks that might be cheap for a good reason, an analyst would most likely apply which of the following criteria?

A. Positive net income

B. Debt-to-total assets ratio below a certain cut-off point
C. Both A and B.

C is correct. Some stocks might be cheap for a good reason such as poor profitability or excessive financial leverage. Hence, the requirement for net income to be positive serves as a check on profitability. Requiring that a company's debt/total assets be below a certain cut-off point would screen out highly leveraged and, therefore, serves as a check on financial risk.